

Business Analytics Report

Synthetic Online Retail Dataset — Customer Behaviour & Sales Analysis

1,000	13	11	5
Total Records	Features	Visualizations	Categories

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SECTION 00

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SECTION 01

Problem Statement

The objective of this analysis is to understand customer purchasing behaviour, product performance, and sales trends using a synthetic online retail dataset of **1,000 transactions**. By applying data wrangling, visualization, and insight generation techniques, the business aims to make data-driven decisions that improve revenue and customer satisfaction.

- Identify top-performing product categories and inventory priorities
- Understand customer demographics — age, gender, and spending patterns
- Analyse price sensitivity and its effect on quantity and reviews
- Evaluate payment method preferences for optimisation
- Discover seasonal sales trends to guide promotional planning

SECTION 02

Dataset Description

The dataset **synthetic_online_retail_data.csv**, sourced from Kaggle, contains transactional and customer-level data spanning multiple product categories, age groups, and payment modes. It comprises **1,000 rows** and **13 original columns**, with one derived feature engineered during analysis.

Category	Column(s)	Description
Customer	age, gender	Buyer demographics
Product	category_name, product_name, price	Item and pricing details
Order	customer_id, product_id, order_date, quantity	Transaction record
Payment	payment_method	Mode of payment used
Feedback	review_score	Customer satisfaction score (1–5)
Derived	total_amount	price x quantity = revenue per order

SECTION 03

Data Wrangling Steps

Step	Action	Detail
Missing Values	Fill — Mean	review_score: 103 nulls filled with mean (3.99)
Missing Values	Fill — Label	gender: nulls replaced with "Unknown" (103 records)
Duplicates	Drop	All duplicate rows removed for data integrity
Data Types	Convert	order_date: object → datetime64 format
New Feature	Engineer	total_amount = quantity × price Range: Rs.20.84 – Rs.2,437.65

Filtering	Explore	Price > Rs.100, gender = M, price between Rs.100–Rs.500
Sorting	Order	Dataset sorted by price in descending order

SECTION 04

Visualizations

Eleven visualizations were created spanning bar charts, pie charts, histograms, scatter plots, and a time-series line plot to provide a comprehensive view of the data.

#	Chart	Type	Key Finding
01	Category Count	Bar Chart	Sports & Outdoors leads — 211 orders out of 1,000
02	Gender Distribution	Pie Chart	Male 45.7% Female 44.0% Unknown 10.3%
03	Payment Method	Pie Chart	Cash on Delivery dominates at 37.4% (374 orders)
04	Price Distribution	Histogram	Mean Rs.251.85 Most products in Rs.100–Rs.400 range
05	Customer Satisfaction	Histogram	Mean rating 3.99/5 75% of customers rated 5 stars
06	Quantity Distribution	Histogram	Mean 3 items/order Range 1–5 items
07	Price vs Quantity	Scatter	No strong correlation — customers buy 1–5 items at all price points
08	Age vs Total Spending	Scatter	Age 30 avg Rs.1,013.60 Age 52 avg Rs.977.25 — top spenders
09	Price vs Review Score	Scatter	No correlation — high price does not guarantee better reviews
10	Quantity vs Total Amount	Scatter	Strong positive relationship — max spend Rs.2,437.65 at qty 5
11	Sales Over Time	Line Plot	Peak: 12 Aug 2024 (Rs.8,788.71) — August is strongest sales month

SECTION 05

Insights

01	Category Analysis	Bar Plot
Sports & Outdoors leads with 211 orders, followed by Electronics (207), Fashion (198), Books & Stationery (193), and Home & Living (191). The narrow gap between categories reflects balanced demand across all segments. ✓ Business Insight: Focus inventory and promotions on Sports & Outdoors and Electronics as they lead in customer demand.		
02	Customer Demographics	Pie Chart — Gender
Male customers account for 457 (45.7%) and Female for 440 (44.0%), while 103 customers (10.3%) carry unknown gender due to missing data. The distribution is essentially equal between male and female. ✓ Business Insight: Design balanced marketing strategies targeting both male and female customers rather than focusing on one group.		
03	Payment Method Analysis	Pie Chart — Payment
Cash on Delivery (374, 37.4%) is the dominant payment method, followed by Bank Transfer (322, 32.2%) and Credit Card (304, 30.4%). Digital methods collectively account for over 62% of transactions. ✓ Business Insight: Ensure smooth COD operations while offering cashback on digital payments to shift customers toward online modes.		
04	Price Distribution	Histogram
Most products fall within the Rs.100–Rs.400 range. Average price is Rs.251.85, minimum Rs.10.72, maximum Rs.499.50. The distribution is fairly uniform, indicating a diverse product portfolio. ✓ Business Insight: Price most products between Rs.100–Rs.400 to match peak customer demand concentration in this bracket.		

05	Quantity Distribution	Histogram
Customers purchase between 1 to 5 items per order, with an average of 3 items . 50% of all customers buy 3 or fewer items per transaction.		
✓ Business Insight: Introduce combo offers or bulk discounts on 4–5 item purchases to increase average order size and transaction revenue.		
06	Customer Satisfaction	Histogram — Review Score
Review scores average at 3.99 out of 5 . 75% of customers gave a score of 5, reflecting high satisfaction. However, ratings as low as 1 signal isolated but significant pain points that require attention.		
✓ Business Insight: Investigate customers who rated 1–2 and proactively offer follow-up support or discount coupons to improve loyalty.		
07	Price vs Quantity	Scatter Plot
The plot reveals no strong relationship between price and quantity. Customers consistently purchase 1–5 items regardless of whether the price is Rs.10 or Rs.499, indicating low price sensitivity on quantity.		
✓ Business Insight: Current pricing structure can be maintained without concern that higher prices will reduce the number of items ordered.		
08	Quantity vs Total Amount	Scatter Plot
A clear positive relationship exists between quantity and total spending. Customers buying 5 items generate totals up to Rs.2,437.65 , compared to single-item buyers at Rs.20.84 — a 117x difference.		
✓ Business Insight: "Buy 3 Get 1 Free" deals or free shipping above a quantity threshold can directly push customers toward higher-volume orders.		
09	Age vs Total Spending	Scatter Plot
Spending spans all age groups (18–75 years). Top spenders: Age 30 (avg Rs.1,013.60) , Age 52 (avg Rs.977.25), Age 29 (avg Rs.947.18). Customers aged 28–35 and 50–55 consistently outspend all other segments.		

✓ **Business Insight:** Allocate higher marketing budgets toward customers aged 28–35 and 50–55 as they generate the greatest revenue per head.

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Price vs Review Score

Scatter Plot

No significant correlation exists between price and review score. Products across the full price spectrum (Rs.10–Rs.499) receive ratings from 1 to 5, with the overall average at **3.99**.

✓ **Business Insight:** Higher pricing does not drive better reviews. Invest in product quality, packaging, and delivery experience to improve satisfaction.

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Sales Over Time

Line Plot

Sales show clear seasonal fluctuations. **Peak: 12 Aug 2024 (Rs.8,788.71)**, followed by 17 Aug 2024 (Rs.7,854.50) and 21 Feb 2025 (Rs.7,425.89). August 2024 is the strongest sales month in the dataset period.

✓ **Business Insight:** Pre-stock inventory before August and launch promotional campaigns to capitalise on peak demand. Use off-peak discounts to stabilise revenue year-round.

SECTION 06

Business Recommendations

01 Inventory	Increase stock for Sports & Outdoors (211) and Electronics (207) — the top two performing categories — to prevent stockouts during demand peaks.
02 Pricing	Keep most products within the Rs.100–Rs.400 range. With an average price of Rs.251.85, this bracket captures peak customer demand and maximises conversion.
03 Payment	Promote UPI and digital wallet payments through cashback and discounts. This reduces Cash on Delivery operational costs and speeds up order fulfilment.
04 Marketing	Run targeted campaigns for the 32–61 age group . Prioritise ages 28–35 and 50–55 who show the highest average spending (Rs.1,013 and Rs.977 respectively).
05 Retention	Customers with review scores below 3 must receive immediate follow-up via email, WhatsApp, or SMS with discount coupons to recover satisfaction and prevent churn.

SECTION 07

Conclusion

This project analyzed the online retail dataset to uncover key patterns in customer behavior and sales trends.

Key Takeaway	Finding	Evidence
■ Most Popular Category	Sports & Outdoors	211 orders — highest across all 5 categories
■ Customer Age Range	32–61 years	Interquartile range of the customer base
■ Average Spend	Rs.737.33 per transaction	Mean of total_amount across 1,000 records
■ Preferred Payment	Cash on Delivery — 37.4%	374 of 1,000 orders used COD
■ Peak Sales Period	August 2024	Rs.8,788.71 on 12 Aug — highest single-day sales
■ Customer Satisfaction	3.99 / 5.0	Good — 75% of customers gave a 5-star rating

The insights generated can directly guide **inventory planning, pricing strategy, targeted marketing, and customer retention efforts** to help the business grow consistently.

— End of Report —